STA2002 Tutorial 6

1. Review

Consider a hypothesis test with **null hypothesis** $H_0(\theta \in \Theta_0)$ versus **alternative hypothesis** $H_1(\theta \in \Theta_1)$. Given a test statistic T and a **rejection region** R, the **power function** $K(\theta)$ of this hypothesis test is

$$K(\theta) = \mathbf{P}(\text{Reject } H_0 \text{ under true parameter } \theta) = \mathbf{P}(T \in R; \theta).$$

Example (One Sample One-sided Test of Mean): Suppose $X_1, ..., X_n \overset{iid.}{\sim} N(\mu, \sigma^2)$, where σ^2 is known. $H_0 : \mu = \mu_0$ vs $H_1 : \mu > \mu_0$. Test statistic is $T = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$. Rejection region $R = \{t : t \geq z_\alpha\}$. The power function is

$$\begin{aligned} K(\mu) &= \mathbf{P}\left(\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \geq z_\alpha; \mu\right) \\ &= \mathbf{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \geq z_\alpha + \frac{\mu_0-\mu}{\sigma/\sqrt{n}}; \mu\right) \\ &= 1 - \Phi\left(z_\alpha + \frac{\mu_0-\mu}{\sigma/\sqrt{n}}\right). \end{aligned}$$

It is noted that $K(\mu_0) = \alpha =$ Type I error. The power function $K(\mu)$ is increasing w.r.t. μ, i.e., if the true $\mu >> \mu_0$, we are more likely to reject H_0.

1. (One Sample Two-sided Test of Mean) Suppose $X_1, ..., X_n \overset{iid.}{\sim} N(\mu, \sigma^2)$, where σ^2 is known. $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$. Test statistic is $T = \frac{|\bar{X}-\mu_0|}{\sigma/\sqrt{n}}$. Rejection region $R = \{t : t \geq z_{\alpha/2}\}$. Please derive the power function $K(\mu)$.

$$
\begin{aligned}
K(\mu) &= \mathbf{P}\left(\frac{|\bar{X}-\mu_0|}{\sigma/\sqrt{n}} \geq z_{\alpha/2};\, \mu\right) \\
&= \mathbf{P}\left(\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \geq z_{\alpha/2};\, \mu\right) + \mathbf{P}\left(\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \leq -z_{\alpha/2};\, \mu\right) \\
&= \mathbf{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \geq z_{\alpha/2} + \frac{\mu_0-\mu}{\sigma/\sqrt{n}};\, \mu\right) + \mathbf{P}\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq -z_{\alpha/2} + \frac{\mu_0-\mu}{\sigma/\sqrt{n}};\, \mu\right) \\
&= 1 - \Phi\left(z_{\alpha/2} + \frac{\mu_0-\mu}{\sigma/\sqrt{n}}\right) + \Phi\left(-z_{\alpha/2} + \frac{\mu_0-\mu}{\sigma/\sqrt{n}}\right) \\
&= \Phi\left(-z_{\alpha/2} + \frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right) + \Phi\left(-z_{\alpha/2} + \frac{\mu_0-\mu}{\sigma/\sqrt{n}}\right).
\end{aligned}
$$

2. Let X equal the yield of alfalfa in tons per acre per year. Assume that X is $N(1.5, 0.09)$. It is hoped that a new fertilizer will increase the average yield. We shall test the null hypothesis $H_0 : \mu = 1.5$ against the alternative hypothesis $H_1 : \mu > 1.5$. Assume that the variance continues to equal $\sigma^2 = 0.09$ with the new fertilizer. Using $\bar{X}$, the mean of a random sample of size n, as the test statistic, we reject H_0 if $\bar{X} \geq c$. Find the minimal n and the corresponding c so that the power function $K(\mu) = \mathbf{P}(\bar{X} \geq c; \mu)$ is such that $\alpha = K(1.5) = 0.05$ and $K(1.7) \geq 0.95$.

Since $\bar{X} \sim N(\mu, 0.09/n)$, we have the power function

$$K(\mu) = \mathbf{P}\left(\frac{\bar{X} - \mu}{0.3/\sqrt{n}} \geq \frac{c - \mu}{0.3/\sqrt{n}}; \mu\right) = 1 - \Phi\left(\frac{c - \mu}{0.3/\sqrt{n}}\right).$$

Given the conditions,

$$K(1.5) = 1 - \Phi\left(\frac{c - 1.5}{0.3/\sqrt{n}}\right) = 0.05,$$
$$K(1.7) = 1 - \Phi\left(\frac{c - 1.7}{0.3/\sqrt{n}}\right) \geq 0.95,$$

we have

$$\frac{c - 1.5}{0.3/\sqrt{n}} = 1.645, \quad \frac{c - 1.7}{0.3/\sqrt{n}} \leq -1.645.$$

Therefore, $n \geq 24.354$. Since n should be an integer, let $n = 25$ and the corresponding $c = 1.5987$.

3. Let Y be $Binomial(192, p)$. We reject $H_0 : p = 0.75$ and accept $H_1 : p > 0.75$ if and only if $Y \geq 152$. Use the normal approximation to determine

   (a) Power function $K(p)$;

   (b) Type I error α;

   (c) Power when $p = 0.80$.

(a)

$$K(p) = \mathbf{P}\left(Y \geq 152;\, p\right) = \mathbf{P}\left(\frac{Y - 192p}{\sqrt{192p(1-p)}} \geq \frac{152 - 192p}{\sqrt{192p(1-p)}};\, p\right) \approx 1 - \Phi\left(\frac{152 - 192p}{\sqrt{192p(1-p)}}\right).$$

(b)

$$\alpha = K(0.75) = 1 - \Phi(1.33) = 0.0918.$$

(c)

$$K(0.8) = 1 - \Phi(-0.289) = \Phi(0.289) = 0.614.$$